



The Evolution of Tea Harvesting: A Comprehensive Review of Machinery and Technological Advancements

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Abstract

Purpose Tea harvesting is a critical stage in the production of high-quality tea, and advancements in machinery and technology have had a significant impact on this process. The conventional manual plucking method for selecting good quality tea buds has faced several challenges, primarily due to labour shortages and rising wages. As a result, the search for mechanical alternatives has been ongoing for the past few decades. This review paper aims to provide a comprehensive analysis of the evolution of tea harvesting, with a specific focus on the machinery and technological advancements that have shaped the tea industry.

Methods In this review, both manual and mechanical tea-plucking methods are comprehensively examined. The evolution of tea machinery, starting from simple clippers to advanced computer vision-based automatic selective tea harvesters (intelligent pluckers), is discussed. Furthermore, this review provides a guide for further improvements in intelligent pluckers.

Results This review paper presents a comprehensive classification of tea-plucking machinery. It examines the evolution of tea plucking and identifies four generations of machinery. The advantages and challenges associated with each type of tea-plucking machinery are thoroughly discussed.

Conclusion Despite the introduction of a few mechanical selective pluckers, their usage is not yet widespread. However, recent rapid technological advancements have created new prospects for the development of intelligent tea pluckers. Currently in the evolving stage, these intelligent pluckers face several challenges concerning plucking efficiency and accuracy. If their design proves suitable for both flat and hilly terrains while remaining affordable for all, there is potential for this intelligent plucking machinery to potentially replace manual plucking in the future.

Keywords Artificial intelligence · Computer vision · Manual plucking · Mechanical shearer · Tea quality

Nomenclature

IoT	Internet of Things
TRI	Tea Research Institute
SL	Sri Lanka
PSTP	Partial selective tea plucker
YOLO	You only look once
DPR	Delta parallel robot

Introduction

Tea (*Camellia sinensis*), one of the most widely consumed beverages in the world, holds a significant place in various cultures and traditions. The annual global tea production is 28.2 million tons (mT). China is the largest tea producer in the world (13.7 million tons; mT) followed by India (5.5 mT), Kenya (2.3 mT), Türkiye (1.5 mT), Sri Lanka (1.3 mT), and Vietnam (1.1 mT) in 2022 (F.A.O, 2024). The cultivation of tea involves meticulous care, from the selection of suitable tea plant varieties to the nurturing of tea gardens under specific climatic conditions (Lu et al., 2022). Tea offers an array of health benefits. It is rich in antioxidants that help combat free radicals, promoting cellular health (Abhiram et al., 2023). Regular tea consumption is associated with improved heart health, reduced risk of stroke, enhanced brain function, boosted metabolism, and potential

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protection against certain types of cancer and chronic diseases (Wang et al., 2021; Xiao et al., 2023).

Tea harvesting or plucking is the primary process before tea processing and it is typically done by hand, where skilled labour is traditionally employed to pluck tea leaves (Abhiram et al., 2020). These skilled labourers perform selective plucking of tender tea leaves, specifically targeting one bud and two leaves, in order to produce high-quality tea. However, the manual harvesting process can be a labour-intensive, time-consuming, and costly operation which accounts for nearly 35% of total production cost worldwide (Wijeratne, 2012; Xu et al., 2022). Although there has been a significant increase in the global demand for tea and tea production has experienced a significant increase (Abhiram et al., 2018; Arhin et al., 2022), the tea sector has been facing difficulties in finding skilled labour to perform plucking (Ranasinghe, 2020). The primary reasons for this can be attributed to the migration of younger generations towards urban areas in search of better jobs, physically demanding nature of tea plucking, comparatively lower wages in comparison to other agricultural sectors, poor living standards, exploitation of human resources, and a lack of social recognition (Ranasinghe & Wickramasinghe, 2021; Saha et al., 2024; Van Der Wal, 2008).

In order to overcome the labour shortage, mechanical shear harvesters were introduced for tea plucking in several countries (Nyasulu, 2001). The evolution of tea harvesting machinery has brought about several benefits including increased efficiency, greater productivity, reduced labour requirements, consistency and precision, improved ergonomics, and scalability (Salvatián et al., 2014; Xu et al., 2022). Although these benefits are acknowledged, it is important to note that most mechanical tea harvesters perform non-selective plucking, making them more suitable for bulk tea production rather than the production of fine-quality tea (Firouzi & Azarian, 2019; Xu et al., 2022). To ensure the production of fine tea, it is necessary to manually or mechanically remove the damaged and coarse leaves from the harvested tea leaves obtained through non-selective plucking methods. Additionally, mechanical harvesters can damage tea leaves, leading to a reduction in tea leaf standards. Moreover, non-selective harvesting removes the maintenance foliage and leads to drastic yield reduction (Wijeratne, 2001). They can also weaken the tea bush, increase the time required for the next tea-plucking cycle, and require skilled operators, which are some of the limitations associated with shear mechanical harvesters (Wijeratne, 2012).

In selective plucking, leaving the immature shoot (arimbu) without harvesting is imperative to get the next plucking cycle on time and get a sustainable yield (Pathirana et al., 2023). Hard plucking is practised during the rainy season whereas light plucking is practised in the dry season in order to sustain the maintenance-foliage (Wijeratne, 2001). The initial selective mechanical tea plucker,

named after its inventor as the “William Selective Plucker”, uses two rollers to mimic traditional selective plucking, targeting only the tender leaves (Williams, 2022). This remarkable innovation has paved the way for enhanced efficiency, consistent quality, and increased sustainability within the tea harvesting process. Subsequently, modifications were made to shear pluckers to convert them into selective pluckers. Following the technological revolution, the tea industry has embraced computer vision, artificial intelligence, and the Internet of Things (IoT) for tea harvesting (Guo et al., 2023). Computer vision automates tea leaf selection based on specific criteria, improving harvesting efficiency. It eliminates human errors, ensuring a consistent and uniform harvest, and enhancing tea quality. This technology reduces labour and time, streamlining operations for faster, more efficient tea production (Xu et al., 2022). Nevertheless, this type of tea plucker is still undergoing research and has not yet been introduced to the market.

A comprehensive review of tea harvest automation is currently unavailable. Therefore, this review aims to provide a thorough overview of the evolution of tea harvesting machinery, exploring the scholarly articles and patent documents. Additionally, this study introduces a new classification for tea harvesting machinery, discusses challenges for widespread adoption, and offers insights into the potential for further advancements in tea harvesting practices.

Tea-Plucking Standards with Tea Quality and Yield

Tea-plucking standards are guidelines that are used to determine the quality and value of tea leaves based on their appearance, size, and other characteristics. The standards can vary depending on the country and region where the tea is produced, as well as the type of tea being produced (Owuor & Obanda, 1998; Owuor et al., 2000). The tea-plucking standards recommended in many countries are fine plucking, medium plucking, and coarse plucking (Han et al., 2014; Shah & Pate, 2016; T.R.I., 2003; Tang et al., 2018). Fine plucking includes a bud with two tender leaves whereas medium plucking includes a bud with three leaves. For coarse plucking, a bud with three or more leaves is harvested.

Several studies confirmed that tea-plucking standard impacts the overall quality of the tea, including flavour, aroma, appearance, texture, and liquor characteristics (Aaqil et al., 2023; Kamunya et al., 2013; Su et al., 2021). According to Owuor et al. (1987), chemical compositions such as catechin, total theaflavins, thearubigins, and sensory properties including brightness, colour, flavour, and aroma were decreased with the coarse plucking of tea leaves. A similar study showed that non-selective plucking significantly reduced theaflavins, total colour, brightness, flavour index, and other sensory parameters compared to selective plucking. Tender leaves are preferred as they contain higher

concentrations of volatile oils and amino acids and lower concentrations of tannins (Owuor et al., 2000). The volatile oil in tea leaves is more associated with flavour and aroma (Guo et al., 2021; Yang et al., 2013). Tender tea leaves are preferred for producing high-quality tea as they contain a lower level of polysaccharide cellulose (Abhiram et al., 2023; Xu et al., 2021). In contrast, matured leaves contain a higher level of polysaccharide cellulose, which contributes to a stalky texture and leads to lower tea grades. Therefore, selective plucking with good plucking standards is essential to ensure fine tea quality.

Classification of Tea Plucking or Harvesting

Tea plucking can be divided into two main categories: manual plucking and mechanical plucking, as shown in Fig. 1. Manual plucking is carried out using either bare hands or a blade attached to a wearable glove. On the other hand, mechanical plucking involves the use of machinery, which can be further classified into selective and non-selective pluckers. Selective pluckers utilise rollers or oscillating blade or employ computer vision technology to selectively identify and pluck tender leaves. Non-selective pluckers, on the other hand, are designed to harvest leaves without specific selection criteria. Batteries and gasoline are the major power sources of mechanical harvesters. These harvesters employ three types of cutting blades such as reciprocating blade, rotary blade, and cylindrical blades. The selection of the blade type is based on the shape of the tea bushes

maintained for plucking. Based on operation, mechanical harvesters can be classified into portable harvester, drivable harvester, and automatic and semi-automatic harvesters. Portable harvesters can be operated by either one person or two people. Mechanical shear pluckers are examples for portable tea harvesters. Tea Garden Tractor-65 and crawler tea harvesting machine are examples of drivable harvesters. Teabot and Valiant tea harvesters belong to automatic tea harvesters. Collaborative tea harvester is a semi-automatic tea harvester.

Manual Plucking

Manual Plucking with Bare Hand

Manual tea plucking is the process of harvesting tea leaves by hand using fingers or a small knife or scissors. This method allows for selective plucking to choose the correct leaf standards for the different types of tea production and also leads to high-quality tea (Firouzi & Azarian, 2019). When manually plucking tea leaves, the plucker carefully chooses the right leaves to harvest based on their size, shape, and maturity. They typically use hands to snap or pull the leaves off the tea plant, taking care not to damage the stems or adjacent leaves (Burgess et al., 2006). The plucked leaves are then collected in a basket or bag and taken to a processing facility for further tea processing. Manual tea plucking is a labour-intensive process and can be more expensive than

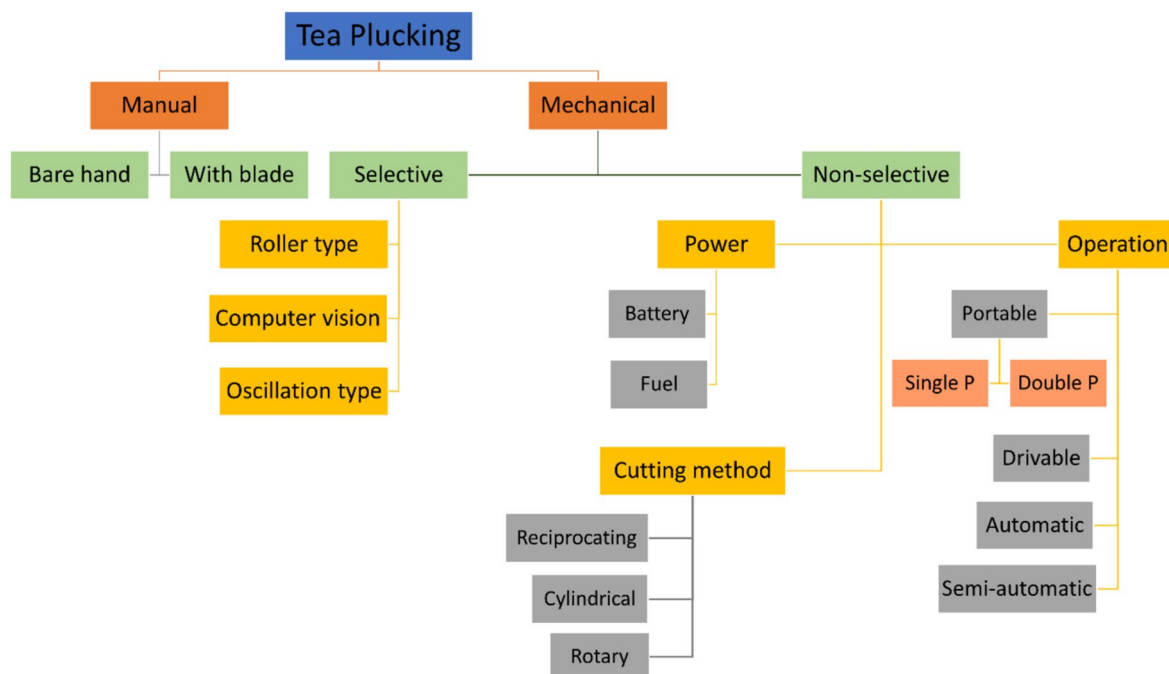


Fig. 1 Different types of tea plucking methods (Single P and double P stand for single person and double persons)

machine plucking (Wijeratne, 2012; Xu et al., 2022). About 70% of the field workforce is dedicated to manual tea plucking, which in turn contributes to approximately 40% of the cost of production (COP) of the final tea product (Wijeratne, 2012). Tea is considered the most labour-intensive crop among the plantation crops.

Engaging in manual tea plucking is a demanding and arduous task due to several factors, including the physical strain it imposes, the necessity to work in adverse weather conditions, the constant threat of pest attacks, the obligation to meet stringent quality standards, and the issue of low wages (Han et al., 2019; Wijeratne, 2012). Tea pluckers are required to spend long hours in the fields, bending down to pick the tea leaves. This process can cause strain and pain in the back, neck, and shoulders. Additionally, plucking requires repetitive hand and finger movements, which can lead to repetitive strain injuries (Supplementary Fig. 1). Also, pluckers carrying tea baskets or collecting bags on their back which causes back and neck pain.

Manual Plucking Aided with Blade

In order to minimise the strain on the figure, few finger and hand protective implements are introduced to reduce the damages to the hand or finger in tea plucking (Supplementary Fig. 2). This particular implement is a glove which protects the hand from injuries and chemical exposure during plucking (Saha et al., 2015). The glove includes a wearable blade attachment that can be inserted into the index finger, with an adjustable width to fit the finger. The attachment has a blade that can be used to cut the tea stem during plucking. Another wearable implement can be inserted into the thumb which has a plastic liner. During plucking, the blade is pressed against the plastic liner to cut the tea stem. This minimises the force required in the pinching or twisting action of conventional plucking thus bruises or abrasion damages can be minimised, if not eliminated.

Mechanical Plucking of Tea Leaves

Mechanical plucking of tea leaves is an alternative to hand plucking, and it uses specially designed machines that move along the tea bushes, plucking the leaves and shoots. Mechanical plucking is typically used for large-scale tea production where manual labour is limited or expensive (Wijeratne & Lanka, 2018). In some countries, mechanical pluckers are used only during the rapid growth season (rush crop) of tea (Pathiranage et al., 2023; Wijeratne, 2003). Three different plucking table shapes or tea bush surfaces are identified; flat, convex or arc, and triangular (Torii, 1966). Therefore, the selection of blade shape depends on the plucking table shape. Straight blades are appropriate for flat plucking tables, such as those found in Sri Lanka. Conversely, in China and Indonesia, concave blades are favoured, as they conform to the curved shape of the plucking table.

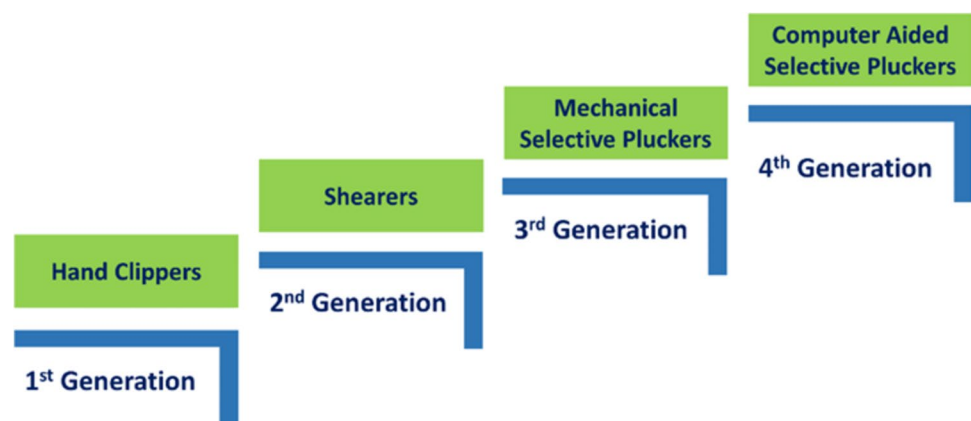
Tea-plucking machinery has undergone several generations of development, each with its own improvements and advancements (Fig. 2). The first generation of tea-plucking machinery consisted of simple hand clippers. The second generation of machinery introduced motorized shearers, which were originally designed for pruning but also used for tea plucking. Both these two generations of machinery performed only non-selective plucking. Selective tea-plucking machinery was only introduced in the early twenty-first century. Third-generation machines were developed to perform selective plucking using rotating rollers or oscillating blades, which could select suitable tea shoots based on their length and strength of the shoot. Fourth-generation tea-plucking machines are currently being developed with the integration of machine vision and artificial intelligence.

Non-selective Mechanical Plucking

Hand Clippers

Hand clippers, which belong to first-generation plucking machinery, are designed with a combination of two blades to

Fig. 2 Different generations of tea-plucking machinery



remove the leaves from the tea plant. The tea shoot is sliced by both blades of the clippers, and the harvested shoots are collected temporarily in a tray attached to the clippers. Once the tray is filled, the harvested shoots are then transferred into a collection bin or bag carried by the plucker (Fig. 3). Perforation on the trays aids to drain the water droplets from the tea leaves. In the initial design, the tea plucker featured a long handle similar to that of scissors, requiring both hands to be moved together in order to perform the plucking action.

Later, the TRI, Sri Lanka, developed a patented hand clippers product with the intervention of P.P.P. Jinadasa (Pvt) Ltd in 1998 (Fig. 3c) (Jinadasa, 2023). In this improved version, the handle was shortened to provide better control during plucking and to apply direct shear force on the tea leaves. The plucker holds one end of the tray, which helps in sensing the plucking table. Furthermore, the blades have a step height from the tray. Both of these facilities enable the plucker to prevent the plucking of immature leaves (*arimbus*). However, it is challenging to avoid plucking over-matured tea leaves since there is no specific mechanism to select the right mature leaves. It was reported that this product increased the plucking out by 50–100% while maintaining the quality of the made tea (Wijeratne, 2012) (Table 1). They were a significant improvement over manual hand plucking, allowing for faster and more efficient tea harvesting. However, they were still limited in their precision and accuracy and were eventually replaced by more advanced tea-plucking machinery. Today, hand clippers are still used in some small-scale tea plantations where labour is abundant and machinery is limited.

These TRI clippers were tested in high-elevation tea plantations in Sri Lanka and a new Great Western Plucking Basket (GWPB) metal made of aluminium bar, and the net was also introduced (Ravi Shankar et al., 2003). The results showed that shear harvesting with shears and the GWPB increased the productivity of medium pluckers by 22% in the morning hours (Table 1). There was no significant difference in the quality of the tea harvested manually and using shears. However, manual harvesting had a 16% higher leaf standard compared to shear harvesting. Using clippers in the morning time increased productivity by 22%. The quality of the made tea from both plucking methods does not show a significant difference. A hand clipper similar in design to the TRI hand clippers was developed and subjected to extensive testing over a duration of 8 years. (Burgess et al., 2006). Although the initial yield of the clipper was not different from manual plucking, a decline in yield was noticed with time (8–14%). Further, there were relatively high proportions of broken shoots (40–48%) and coarse material (4–6%) compared to manual plucking.

A similar design was made by Tirthankar Ray and Nalavade (2013), and it was motorised to increase the harvesting speed (Supplementary Fig. 3). A DC gear motor with 100 rpm and 0.9 Nm torque was attached to the handle allowing it to move the blade and perform the clipping action (Table 1). The motor was powered with a 12 V and 7 Ah lead acid battery. It was reported that the field coverage rate was 0.015 ha h⁻¹ which was nearly 3 times higher than manual plucking. The plucking rate was 8 kg h⁻¹ and it was nearly 2.5 times higher than manual plucking.

Fig. 3 Hand clippers used for tea plucking; **a** scissor type clippers, **b** tea plucking using scissor type clippers, **c** TRI, SL developed clippers, and **d** tea plucking using TRI, SL developed clippers (Jinadasa, 2023)



Table 1 Non-selective mechanical tea harvesters (N/A means not available)

Harvester type and name	Field coverage rate	Plucking rate	Annual yield (kg/ha)	Coarse leaf %	Power source	Important finding/s	Reference
Hand clipper (TRI)	N/A	4.3 kg/h	N/A	N/A	N/A	Using clippers in the morning time increased productivity Clippers increased productivity by 22% than manual plucking (MP)	Ravi Shankar et al. (2003)
Hand clipper	N/A	N/A	5470	4–6	N/A	Broken and coarse leaves were significantly higher MP Over 7-year period, harvest declined than MP	Burgess et al. (2006)
Motorised hand clippers	0.015 ha/hr	8 kg/h	N/A	N/A	12 V, 7 Ah battery	Plucking rate was 2.5 times and field coverage was 3 times higher than MP	Tirthankar Ray and Nalavade (2013)
Forbes & Walker harvester with helically arranged picking arms (BatFW),	0.17 ha/day	N/A	2086.5	7.13	Battery	Higher damage level for tea leaves and lowering the tea leaf standard	Pathirana et al. (2023)
Battery-operated Kawasaki harvester with reciprocating blades (BatKW),	0.25 ha/day	N/A	1974.9	14.84	Battery	Remove more dormant bud is good, but removing excessive immature leaves reduces yield and frame development	Pathirana et al. (2023)
Petrol-driven Kawasaki harvester with reciprocating blades (PetKW)	0.47 ha/day	N/A	2297.2	11.67	Petrol	The larger the machine, the higher the damage to harvested leaves and the lower the girth of the secondary branch	Pathirana et al. (2023)

Portable Tea Harvester

Portable Shear Blade Type Tea Harvester

The battery-operated tea shearers are more portable and easier to manoeuvre than traditional tea shearers due to their low weight (Fig. 4). They are environmentally friendly as they produce minimum noise level and does not emit harmful fumes and gases. However, most of the battery-operated tea shearers in the market cannot be used for a long time continuously due to their poor ergonomic design, especially the weight of the motor also acts on the hand and poor weight distribution. Holding these shearers in a single hand for long hours can create fatigue and stress on the hands (Wijeratne, 2012). Further, their plucking capacity is very low (20–30 kg/h) compared to its counterpart (500–600 kg/h), fuel-operated one. The bag containing a battery is designed to be carried on the shoulder. Typically, battery-powered tea shearers can be used continuously for 3 to 5 h after a single charge (Kakati, 2017). Generally, tea shearers employ a reciprocating type blade.

The fuel or gasoline-operated tea shearers are powered by fuel or gasoline engines, making them suitable for use

in areas without access to electricity. They are designed to be lightweight, portable, and easy to operate, which allows workers to use them for extended periods without fatigue. These machineries perform non-selective plucking and are thus suitable for low-grade tea manufacturing.

A typical fuel-operated tea shearer is shown in Fig. 5. This has two main components; engine and cutting component. Both two-stroke and four-stroke engines are used for tea shearer fabrication, but two-stroke engines are very common. Generally, 0.7–1.5 HP engines are used in the tea shearers. An engine-mounted shoulder strap is utilised to suspend the tea shearer on the operator's back. Although effective, these machines are known for their high noise levels and fume emissions, necessitating the use of earplugs to mitigate the noise's adverse effects. The cutting components have blades, handle, gear, tea collecting bag, blower, and blower pipes. Blades are available in varying widths to match different plucking table sizes. Since this machine is heavier than battery operated counterpart, two handles are given for easy manoeuvre of the cutting component (Kakati & Das, 2016). The gear is used to change the speed of the cutting blades. The blower and blower pipes aid in inflating the tea collection bag and propelling the sheared tea leaves

Fig. 4 The battery-operated tea shearer; **a** components and **b** application in the field (BOMA, 2024)



Fig. 5 Fuel-operated tea shearer for tea plucking (Hayles, 2024a)



into the bag. The plucking capacity of this machine is several times higher than the battery-operated one which ranges between 0.7 and 1.0 ha day (Hayles, 2024b).

Pathirana et al. (2023) conducted a comprehensive study to compare locally available three motorised harvesters in Sri Lanka: Forbes & Walker harvester with helically arranged picking arms (BatFW), battery-operated Kawasaki harvester with reciprocating blades (BatKW) and petrol-driven Kawasaki harvester with reciprocating blades (PetKW), against manual plucking (MP) (Table 1). Reciprocating blades (BatKW & PetKW) sliced all the shoots irrespective of leaf grades whereas helical arms (BatFW) left random shoots in the plucking table. The presence of random leaves, including harvestable tender leaves, necessitated the need for the plucking machine to pass over the tea bush once more for thorough plucking. This resulted in a decrease in labour efficiency. BatKW gave significantly ($P < 0.05$) highest coarse leaf percentage (14.84%) followed by PetKW (11.67%) and BatFW. The good tea leaf percentages significantly ($P < 0.05$) differed between machinery and they were in the following order: MP (94%) > BatKW (80%) > PetKW (75%) > BatFW (60%). The harvested leaves stuck between rotary arms and blades thus they caused more damages to leaves than reciprocating blades. This damage level and randomly left over leaf percentage in plucking table were increased with operational speed. BatFW left higher level of immature leaves and dormant shoots in plucking table compared to other machinery. Further, the study reported that girth of secondary branches and frame development were poor for machine harvesting. The Larger the machine, the poorer the frame development. This could be attributed to the severity of the plucking which directly influences the carbohydrate translocation from leaf to roots. Root starch level did not significantly differ between plucking methods until 6 months, but significantly ($P < 0.05$) decreased for mechanical harvest at 12-month period. As a result, all the motorised harvesters significantly ($P < 0.05$) decreased the made tea yield by 46–51% compared to manual harvesting. Therefore, this study suggested that manual plucking is beneficial in all aspects with respect to mechanical plucking. If mechanical plucking is unavoidable, using reciprocating blades type harvesters ensure better yield and maintenance foliage. Nevertheless, the design of BatFW needs to be improved to minimise the damage level for tea leaves.

Kakati and Das (2016) developed a modified harvester with helically arranged picking and tested in Assam, India. In this designed, the cutting blade width increased from 196 to 510 mm. As a result, the weight of the harvester was subsequently increased, leading to the separation of the battery from the harvester and its relocation to the operator's back. Additionally, the harvester was equipped with a waist belt to provide support to the operator's body while handling

it. This facilitated easier manoeuvring of the cutting blades over the tea bushes. Furthermore, a lever was attached to the harvester, allowing for precise control and movement of the cutting blades. The grip strength of the workers was tested before and after manual and mechanical plucking. It was found no significant differences between both plucking methods. The ergonomic was tested and found that workers with waist height below 800 mm felt on their arms and workers with waist height felt hip pain.

The cutter shape and other specifications have a significant influence on the damage level to shoot, harvest quality, cutting speed, and power consumption of the harvester (Li et al., 2011). Therefore, a study was conducted to optimise the reciprocating cutting system of a single-row tea harvester (Du et al., 2022). This study proposed the use of bionic blades and cutters inspired by the curved structural characteristics of the upper jaw of crickets. This study investigated the effects of inter-node number, tea variety, and blade type on cutting force and power consumption. The results showed that the bionic blade reduced cutting resistance and power consumption, while tea variety had minimal effects on these factors. The optimum specifications of the bionic blade were as follows: cutting speed of 0.8 m/s, a forward speed of 1.0 m/s, a cutting angle of -3° . This parameter combination resulted in an integrity rate of 92.7% and a missing rate of 3.9% for the tea shoot.

Surf Type Portable Non-selective Tea Harvester

Wheeled or tracked tea harvesters face challenges including limited manoeuvrability on uneven terrain, soil compaction from their weight, potential damage to tea bushes during operation, and the need for regular maintenance and repair due to complex mechanical systems. Further, additional space is required for the space for manoeuvring the harvester which decreases planting capacity of the field (Williames, 1992). To address the issues associated with wheeled or tracked tea harvesters, Williames Hi-Tech International Pty. Ltd developed a prototype known as the “T1000 Magic Carpet” which has been patented (Williames, 1992). This innovative harvester eliminates ground contact by forgoing wheels or tracks (Supplementary Fig. 5). Instead, it utilises a surfing mechanism that allows it to glide directly on top of the tea bushes (Williames, 2000). This was tested in different countries including India, South Africa, Malaysia, and Papua New Guinea. This harvester is claimed to be more precise and increases the density per area of tea plant by 15% (Williames, 2000). However, a few shortcomings need to be acknowledged. Firstly, it cannot be used in steep or undulating terrain due to its surfing mechanism. Secondly, there is a risk of potential damage when it is used on young tea shoots.

Drivable Non-selective Tea Harvesters

Tea Garden Tractor

The drivable tea harvester has a long history. In early 1960s, Japan started to investigate drivable tea harvesting machinery. The Tea Experiment Station of Kagoshima Prefecture has developed a tractor called the Tea Garden Tractor-65, which is designed for general use in tea garden management. This tractor is equipped with a plucking unit and can be used in tea gardens with a row width of 1.8 m. The plucking unit consists of an arc-shaped cutter with a radius of curvature of 1.0 m and a plucking width of 1.72 m. It operates in a reciprocating manner and can effectively pluck tea leaves from plants ranging in height from 0.5 to 1.0 m. The plucked leaves are collected through suction into two leaf-tanks with a total volume of 1.3 m³ (Nakano, 1976). Later, a small-sized crawler-type tea-plucking machine (commonly referred to as Matsumoto type) has been developed with the aim of improving efficiency and reducing soil compaction in the narrow space between hedges. This machine is designed to pluck one row of tea plants in two strokes, making it a half-row type. Two operators operate this harvester at efficiency of 10 acre/h (Nakano, 1976).

Crawler Tea Harvesting Machine

The Japanese automatic tea harvesters are heavy machines that require a larger inter-row spacing in order to operate effectively in the field. However, many old tea plantations, including those in China, do not have dedicated paths suitable for this type of larger machinery. In order to overcome this, Li et al. (2024) developed a crawler tea harvester that is smaller in size and can operate in hilly areas and narrow inter-row tea fields. The machine is equipped with a tracked chassis, a picking unit, a collection unit, a tea canopy sensing unit, and a cutter height adaptive adjustment unit (Supplementary Fig. 4). The cutter height adaptive adjustment unit utilises a sensing plate with limit sensors to automatically adjust the height of the cutter. The optimised parameters for the machine were determined to be a machine speed of 3.170 km/h, crank radius of 18.737 mm, average cutter speed of 0.951 m/s, and a processable tea rate of 91.38%. Field trials were conducted to validate these parameters, and the results showed that the processable tea rate achieved was 91.26%, which closely aligned with the optimised results.

Automatic Non-selective Tea Harvester

Valiant Tea Harvester

Matsumoto Kiko Co., Ltd., a company based in Japan, has developed a tea harvester that offers the flexibility of being

both self-propelled and drivable (Fig. 6a, b) (New Century Corporation, 2018). This is an example of an automated shear plucker. An arc-type blade on the upper side moves against the bed bar blade on the lower side, performing non-selective plucking. These blades have a length ranging from 2250 to 750 mm. These harvesters have the capability to harvest approximately 4000 to 5000 kg of tea leaves and can cover an area of 4 to 5 hectares within 6 h. The harvested tea leaves are collected in a basket attached to the harvester, with a capacity of 800 to 1000 kg. In autonomous mode, two ultrasonic sensors guide the harvester along the middle or centre position of the tea plant row. While the harvester is in the centre position, both sensors receive equal reflectance from the tea bush (Fig. 6c). However, if the harvester is moved from its centre position, the sensors detect the unequal reflectance (Fig. 6d). The harvester then automatically adjusts its position until it reaches equal reflectance, effectively bringing it back to its initial centre position. Once a row is fully harvested, the harvester automatically moves and aligns itself with the next row, aided by position sensors. When the harvesting process is completed, the program initiates an automatic shutdown of the harvester.

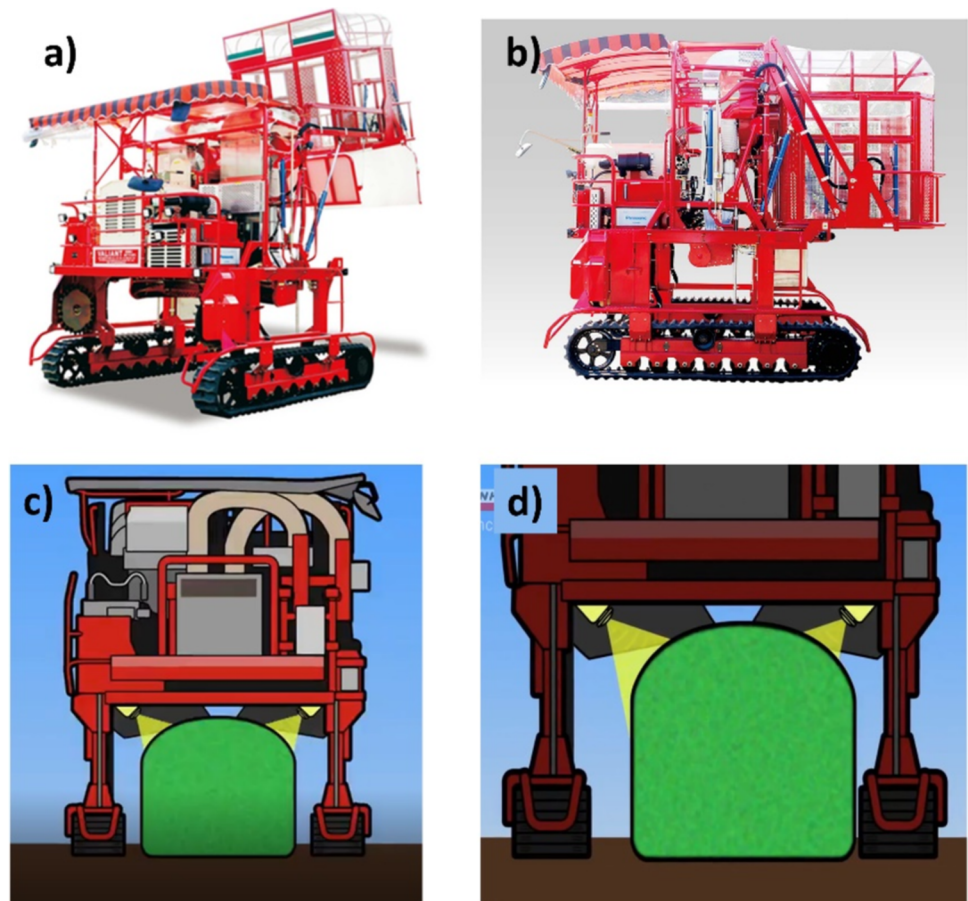
Teabot-Robotic Non-selective Harvester

Tregothnan Tea, the exclusive tea producers in England, developed a solar-powered robotic tea harvester called the “Teabot” (Cullinane, 2023). This remote-controlled harvester operates on tracks, making it suitable for muddy soil conditions. The harvesting system of the Teabot resembles that of the Kawasaki motorized harvester, utilizing moving shear blades to cut the tea leaves. However, it is important to note that the harvesting process is non-selective. To collect the harvested tea leaves, a tray is positioned beneath the solar panel on the machine.

Semi-automatic Tea Harvester Machinery

In Taiwanese tea plantations, self-propelled harvesting machinery is not suitable, hence shear pluckers operated by two people are commonly used. However, during peak seasons with limited labour, Lai et al. (2022) developed a collaborative robot for tea harvesting. This robot replaces one labour and works under the guidance of another person. It is an example of semi-automated tea shearer. The robot is controlled with adjustable autonomy, allowing the human to supervise and adjust the cutting tool's height and speed. The robot's relative motion is controlled using a marker tracking method. The operator wears a special marker (AR tags) that is tracked by the robot's RGBD camera. The developed method for tea harvesting with the collaborative robot has proven to be effective and feasible. It improved harvesting quality, reduces the workforce by up to 50%, and minimised physical strain for farmers.

Fig. 6 Valiant tea harvester: **a** full view, **b** side view, **c** cantered position, and **d** off-cantered position (New Century Corporation, 2018)



Challenges of Non-selective Mechanical Plucking and Strategies for Improvement

In tea harvesting, the sink-source relationship is a crucial aspect that impacts the quality and yield of the tea leaves. This relationship refers to the balance between the sink, which represents the developing leaves, and the source, which represents the photosynthetic capacity of the tea plant. The source produces sugars through photosynthesis and transports them to the developing leaves to support their growth. During the initial growth phase of tea shoots after harvest, they function as sinks. As they reach the appropriate maturity, they transition into sources (Wijeratne, 2003). When harvesting takes place, these fully matured leaves are removed, reducing the source capacity of the tea bushes. This is why mother leaves, which act as the source and are known as maintenance foliage, are important in mother leaf plucking. However, in non-selective mechanical plucking methods, this crucial maintenance foliage is harvested, disrupting the sink-source relationship (Pathiranage et al., 2023). While mechanical plucking may initially appear to provide a high yield, over time, a significant decline in yield becomes evident. This decline is primarily attributed to the prolonged stress caused by an imbalance in the source-sink

relationship. As a result, the storage of carbohydrates in the roots further decreases.

On the other hand, non-selective mechanical plucking involves harvesting immature leaves (arimbus) that would have served as the harvest for the next plucking cycle. The removal of these immature leaves, which act as sinks, diminishes photo-assimilation and disrupts the source-sink relationship. Consequently, the development of the plucking table frame is affected (Wijeratne, 2003). To ensure a healthy sink-source relationship during tea harvesting, it is important to strike a balance between the amount of leaves being harvested and the plant's ability to photosynthesize and produce enough nutrients. Furthermore, non-selective mechanical plucking results in a higher number of dormant shoots remaining on the plucking table. This leads to a decrease in overall yield over time (Pathiranage et al., 2023). The girth of the secondary branches indicates the frame and vascular system development. Non-selective mechanical plucking significantly decreased the girth than manual plucking (Pathiranage et al., 2023).

In tea plantations where labour is expensive or in short supply, mechanical harvesting becomes necessary. However, precautions must be taken when using non-selective mechanical harvesters. To avoid plucking immature leaves, it is recommended to avoid moving the harvester too deep

into the plucking table. Minimising damage to the tea leaves and shoots can be achieved by avoiding repetitive harvesting at the same point (Pathiranage et al., 2023). To provide sufficient space for the movement of mechanical pluckers, it is advisable to follow the spacing recommendations provided by TRI, SL ($2 \times 2 \times 5$ ft) (T.R.I., 2019). Different tea cultivars exhibit varying suitability for mechanical harvesting. Generally, tea cultivars with smaller and erect leaves are more suitable as they incur lower levels of damage during the process. To reduce stress on the tea plants, it is recommended to only use mechanical harvesting during the rapid growth season of tea shoots or to employ mechanical harvesters intermittently (Pathiranage et al., 2023; Wijeratne, 2003). By implementing these strategies, the quality and yield of tea can be effectively maintained. Nevertheless, the development of a highly sophisticated selective-mechanical plucking machinery emerges as the definitive remedy to effectively address the various obstacles encountered by non-selective pluckers.

Selective Mechanical Pluckers

Selective mechanical pluckers are specifically engineered with advanced mechanisms to emulate the manual selective plucking process by exclusively targeting and harvesting the appropriate mature tea shoots. These pluckers belong to the third and fourth generations of plucking machinery, each employing distinct approaches to achieve the desired selectivity. The third-generation machines utilise rotating rollers or reciprocating blades, designed to discern the correct shoots based on their length and thickness. Conversely, the fourth-generation machines employ machine vision technology to differentiate tea leaves based on various phenotypic characteristics, including shape, size, colour, texture, and angle. As a result, the fourth-generation machinery exhibits enhanced accuracy in terms of selectivity.

A Partial Selective Tea Plucker (PSTP)

A partial selective tea plucker (PSTP) was proposed by Piya-thissa et al. (2015). The machine has a weight of 6.5 kg and plucking width was 600 mm. In this design, a shoot selector was placed in front of the cutter blade of the harvester (Supplementary Fig. 6). The shoot selector of the machine had a gap between its teeth that matched the average thickness (5 mm) of a suitable shoot. Any shoot thicker than this average would be prevented from entering the blade. While this mechanism effectively prevents fish leaf plucking, it still allows maintenance foliage (immature shoots) with lower shoot thickness to enter the cutter. As a result, this mechanism is not entirely effective in preventing the harvesting of immature leaves. Although referred to as a selective plucker in the study, it is more accurately described

as a partial selective plucker. This design was tested against manual plucking (MP), TRI clipper, and Kawasaki plucker (KW) at TRI, Sri Lanka. It was reported that percentage of stranded leaves (right matured) was in the following order: MP (96.7%) > PSTP (93.4%) > TRI (94%) > KW (88.2%). Non-selective plucking percentage (immature or over-matured leaves) was higher for KW (11.9%), followed by PSTP (6.6%), TRI (6%), and MP (3.4%).

Willames Tea Selective Plucker

The invention of a selective tea plucker is a ground-breaking discovery by the Willames Tea Company. It is a selective mechanical plucking machine that employs a set of reciprocating rollers to pluck only the tender tea leaves (Supplementary Fig. 7). One of the key features of the Willames Tea selective plucker is its ability to adjust to different tea bushes and terrains. The machine is designed to be highly manoeuvrable, allowing it to navigate steep hills and narrow rows with ease (Willames, 2022). It is powered by a diesel engine and is available in a range of sizes to suit different tea farm requirements.

The first design was tested in 2017 which is a light weight plucker, surf on top of the tea bushes with the support of two operators (Supplementary Fig. 7a). These operators push forward the machine using the long handle attached to it and the whole weight of the machine is supported by the tea bush. After modification, it was launched in 2021 to the market and it was a little bigger in size compared to the prototype. Although the earlier design is suitable for uneven terrain, the latest design with wheels is unsuitable for such terrain (Willames, 2000, 2022).

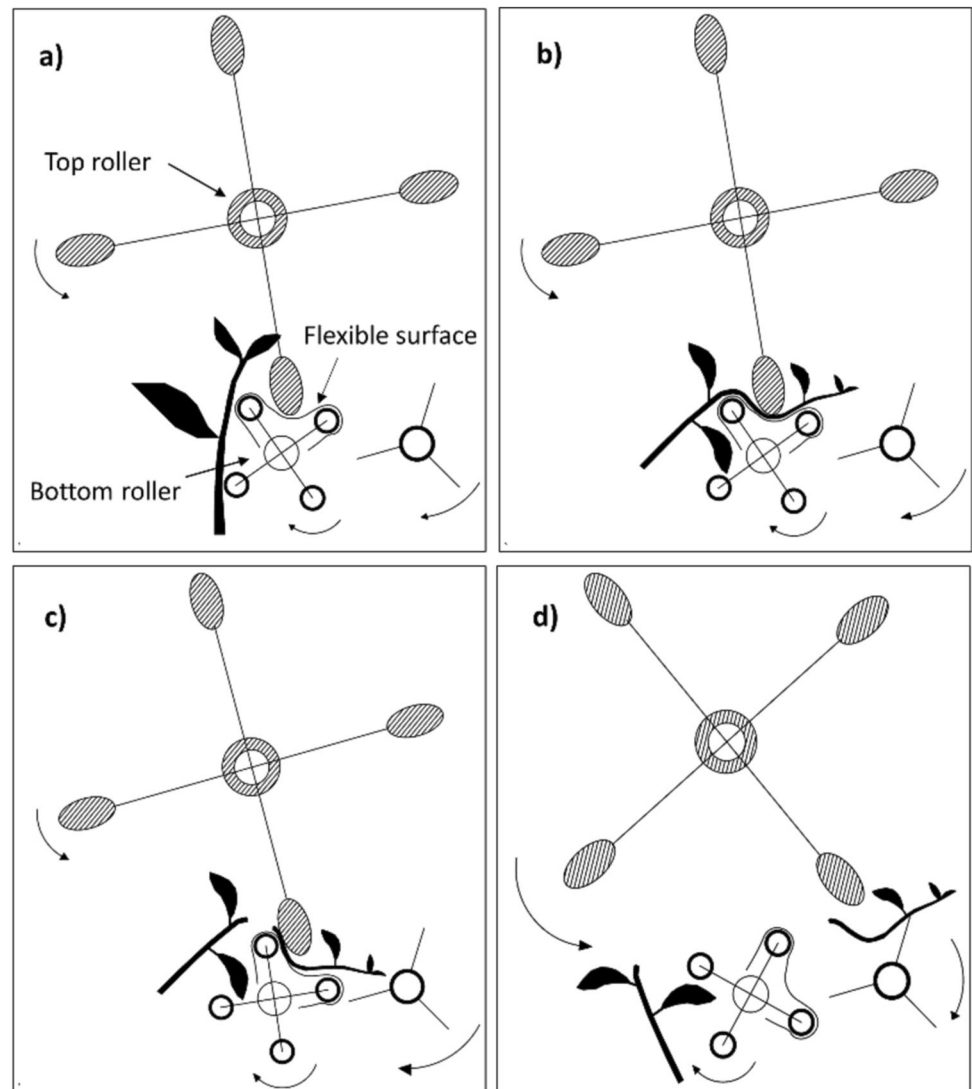
This design has two rollers; one assembled on top (top roller) and another one is assembled a little offset and bottom (bottom roller) to the top one (Willames, 2022) (Fig. 7). The top roller is larger in diameter than the bottom one. The larger diameter is intended to create more gaps between the arms, which facilitate capture of incoming tea shoot. The bottom roller provides the tea-breaking surface. It is in diamond shape to allow one plant engagement surface to contact and deform the other plant engagement surface. A cylinder in bottom roller creates the surface that breaks the plant, and to ensure a potential breaking point or area is formed at the bend of the stem. To achieve this, the diameter of that cylinder must be small enough to create a tight radius. The bottom cylinder surface must have sufficient friction to grip the tea shoot without slippage. Further, the surface of the roller is deformable to bend and twist the tea shoot. Therefore, a textile or a rubber belt or liner is more suitable for this purpose. Both these rollers revolving in opposite direction. Of which, top roller rotates anticlockwise and bottom roller in clockwise direction (Willames, 2022).

The mechanism of selective plucking is shown in Fig. 7. Top roller catches the tea shoot and pushes it towards to the bottom roller (Fig. 7a). Once the tea shoot is trapped in the gap between the two rollers, the top roller presses the shoot against the second roller (Fig. 7b). As both rollers revolve further in opposite direction, high tension is created on the tea shoot. When the tension exceeds the tensile strength of the shoot, it snaps (Fig. 7c). In the subsequent rotation of the top roller, the tea shoot is thrown away (Fig. 7d).

The selective plucking machine (Supplementary Fig. 8) has been intelligently engineered to execute precise selective plucking, and its design incorporates several key features to enable this functionality. Firstly, the length of the rollers plays a vital role in ensuring the proper feeding of shoots with the correct length. Shoots that are too short or too long may fail to effectively grasp by the rollers. Moreover, the centre-to-centre distance between the rollers is carefully

calibrated to maintain an optimal gap between the plucking surface of the bottom roller and the top roller arm. If this clearance is too narrow, it may result in the unintended harvesting of small and medium-length shoots. Conversely, if the clearance is too wide, no shoots will be captured by the rollers. Additionally, the spacing between the pressure points is a critical factor in achieving efficient tea leaf plucking. Two pressure points, namely P1 and P2, are involved in this process. P1 ensures that the tea shoot remains securely in place due to the friction between surfaces, while P2 applies tension to the shoot. The distance between these pressure points serves to determine the appropriate shoot length to be plucked and the corresponding force required. Longer shoots, which possess higher lignin content, necessitate a greater tension force to break. Therefore, the given design may not provide this force as it can only generate a certain maximum force.

Fig. 7 The mechanism of tea plucking of Williams Tea selective plucker; **a** rollers approaching tea shoot, **b** tea shoots traps by the rollers, **c** tea shoot snapped, and **d** tea shoots left the roller (adopted from Williams (2022))



Myers Oscillating Selective Tea Harvester

In 1970, a prototype of an oscillating type tea harvester was developed and subsequently patented (Myers, 1970). The harvesting component consists of two primary parts: oscillating blades and stationary U-shaped rods (Fig. 8). The middle segment of the oscillating blade is constructed with a metal bar and encased in a rubber sleeve. This metal bar oscillates horizontally between the U-shaped rods on both sides. As the tea harvester moves forward, tea shoots become entrapped within the rods and blades. These shoots experience a bending force and snap or fracture at the point where the maximum bending force is applied. This mechanism facilitates the selection of the appropriate mature shoots consisting of one bud and two leaves. Two critical factors contribute to the selection of the correct mature shoots. Firstly, the proper engagement of the blade and U-shaped rod is crucial to apply the necessary bending force to the shoot. Secondly, the flexibility of the rubber determines the force exerted on the shoot. Tender shoots require a greater bending angle to snap compared to mature shoots. Therefore, when the blades capture too tender shoots, they only bend to a lower angle, which fails to break the shoots. Conversely, coarse shoots possess more strength and consequently demand a greater force to break them than the force generated by the aforementioned mechanism.

Computer Vision-Based Automatic Selective Tea Harvesting

In recent years, researchers have shown increasing interest in the application of computer vision and artificial intelligence (AI) for tea bud identification. Numerous machine learning and deep learning techniques have been utilised to accurately identify tea buds and detect their harvesting points along the shoot. The process of tea bud recognition proves to be challenging due to factors such as complex background, varying light conditions, occlusion, shade, and the dense growth of tea plants (Xu et al., 2022). Nevertheless, certain advanced deep learning techniques and ensemble methods have achieved remarkable success in accurately identifying tea buds, with accuracy rates surpassing 90%. However, the development of a comprehensive automated computer vision-based tea harvester is still in its early stages, with only one such system documented in the existing literature (Li et al., 2023). The main sequential functions of an automated tea harvester are shown in Fig. 9. The process commences with the image acquisition system capturing images of the tea-plucking table. Subsequently, the control system processes the acquired image, enabling the determination of the location of tea buds and

the plucking point. Following this, an analysis of the arm path is conducted to ascertain a safe height, and the order of plucking points is established. Leveraging these calculations, the arm is then manoeuvred to execute the tea-plucking operation.

Tea Bud Identification and Picking Point Detection

Tea bud identification and locating the picking point is the first challenge in computer vision-based harvesting systems. The challenges associated with task are complex background, shade, different contrast images, and occlusion (Guo et al., 2023). Computer vision technology, coupled with intelligent algorithms, is being used to accurately identify and distinguish tea buds from other plant parts. These methods analyse visual characteristics of tea buds, such as colour, shape, texture, and size, to differentiate them from leaves or other plant components (S. Zhang et al., 2023a, 2023b, 2023c). For tea harvesting, the first step involves detecting the tea buds, and subsequently, the precise localization of the plucking points is required. Computer vision systems can guide harvesting robots or machinery to selectively pick the desired tea buds, ensuring the production of high-quality tea.

Deep learning techniques have gained significant popularity and widespread adoption in the field of tea bud recognition and plucking point detection, surpassing traditional machine learning approaches (Gunaratnam et al., 2024). Compared to using a single machine learning method, the combination of different methods significantly increased the accuracy of tea bud detection. For example, Xu et al. (2022) combine the fast detection capability of YOLOv3 and high precision capability of DenseNet201 to accurately detect tea buds (Table 2). In this approach, firstly, tea buds were classified using YOLOv3 and then these classified images were inputted into DenseNet201 to remove the false recognition as tea buds in order to increase the accuracy. YOLOv3 alone provided 71.32% of accuracy whereas combination with DenseNet201 gave 95.71% accuracy. Further, this study demonstrated that tea bud images taken from side view gave higher precision than top view images. Chen and Chen (2020) utilised the Faster R-CNN (region-based convolutional neural network) to detect the regions of interest corresponding to the “one tip with two leaves” (OTTL) areas of TTES 8 and TTES 18 tea varieties (Table 2). Subsequently, a fully convolutional network (FCN) was trained to precisely locate the plucking points within the OTTL regions. The Faster R-CNN model achieved a precision of 79% and a recall of 90%, while the FCN achieved a mean accuracy of 84.91% and a mean intersection-over-union of 70.72%. This method was validated using two new datasets from Chin Shin Oolong (CSO) and Shy Jin Chuen (SJC) tea varieties, which were not used for testing purposes. The high

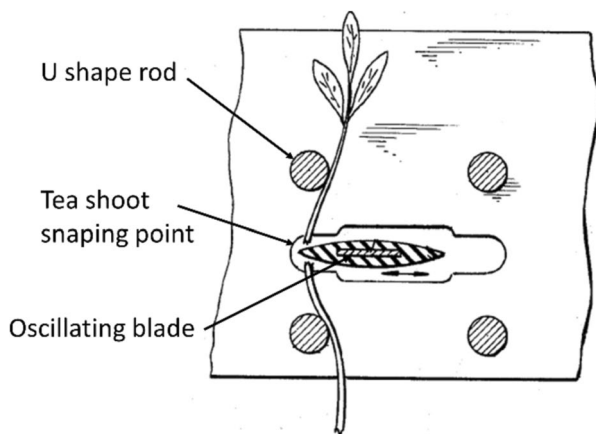


Fig. 8 Schematic diagram of a mature shoot harvesting using an oscillating type tea harvester (Myers, 1970)

precision in tea bud detection demonstrated the versatility of the model, as it can be applied to different tea bush varieties beyond those used for training the models.

In a study, Wang et al. (2023) used mask R-CNN for bud detection and pick the harvesting point in the stem. The approach involved extracting features of tea buds and leaves using the Resnet50 residual network and a feature pyramid network (FPN) (Table 2). Classification and preselection box regression training are performed on the feature maps using a regional proposal network (RPN). The regional feature aggregation method (RoIAlign) was used to eliminate quantization errors, and the output module of the model performs classification, regression, and segmentation functions. The accuracy of leaf and stem detection was 95.53% and 92.37%, respectively. The overall accuracy of the model was 93.95%. The lower accuracy in picking point detection is due to misclassifying regions with similar shape and colour as stems, resulting in false positives. Additionally, stems lack a distinct shape compared to leaves, posing challenges in accurately distinguishing them from surrounding elements. A similar study used Mask R-CNN for bud identification and picking point detection (Yan et al., 2022). The precision of picking point detection (94.9%) was higher than that of bud

identification (44.9%) (Table 2). While the precision of this model was comparable to other tested models such as YOLOv3 and Faster R-CNN for segmentation, it exhibited superior performance in dealing with bud occlusion.

Only limited study used depth camera for tea bud detection. For example, Guo et al. (2023) utilised a segmentation approach based on colour and depth data from a stereo vision camera to accurately detect the shapes of tea buds in 2D and 3D spaces (Table 2). The YOLOv4 deep learning model is modified to create a lightweight model (YOLOv4-lighted) with shorter inference time, and additional attention mechanisms are incorporated to enhance the detection accuracy of tea features. The best model was YOLOv4-lighted + Improved Convolutional Block Attention Module (ICBAM) model which gave 94.19% average accuracy in bud detection. The same model showed 87.1% accuracy and 0.12 positioning time of the picking point. In a study, Meng et al. (2023) proposed an improved version of the YOLOv7 algorithm for tea bud identification. The improved algorithm incorporates Depth Separable Convolution (DS Conv) blocks, Convolutional Block Attention Modules (CBAM), and Coordinate Attention (CA) modules. These enhancements result in improved mean average precision and mean recall rates, achieving 96.70% and 93.88% respectively.

These studies have demonstrated the effectiveness of artificial intelligence techniques, particularly deep learning methods, in accurately recognizing tea buds and determining the plucking points. However, when applying these techniques in real-time applications, several challenges may arise like wind and different light levels. Further studies should also explore the development of robust algorithms that can handle the challenges posed by wind-induced movement and varying light conditions.

Plucking Manipulator and End-Effector Design for Tea Harvesting

A plucking system of a tea harvester comprises plucking manipulator, end-effector, and computer vision (Camera). Plucking manipulator is built to mimic the actions of a human hand,

Fig. 9 The sequential steps of an automatic tea harvester

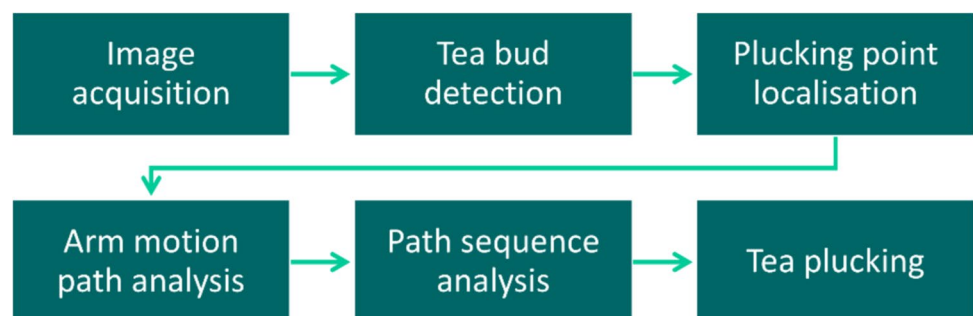


Table 2 Computer vision and artificial intelligence (AI) based tea bud identification (BI) and picking point detection (PPD) (N/A means not available)

Application	Image acquisition tool	Image size	AI method	Performance (precision, F1 score)	Main finding	Limitation/s	Reference
BI	Sony HDR-XR350V; Canon EOS 1200D	Side: 140 Top: 679	YOLOv3, DenseNet201	95.71%, 96.84%	Detection accuracy was high for side images of tea leaves than top view images	Increasing batch size decreases the precision Lighting condition and angle of imaging impact the results	Xu et al. (2022)
BI and PPD	Lumix DC-GF9, Panasonic; M3 EOS, Canon	150	Faster-RCNN, FCN	Faster-RCNN: 79%, NA FCN: 89.47%, NA	This model is species independent	N/A	Chen and Chen (2020)
BI and PPD	Smart phone	1400	Mask R-CNN	93.95%, NA	The leaves identify better than the stem (tea-picking point)	Regions that have similar colour and shape to the stems are misidentified	Wang et al. (2023)
BI and PPD	Smart phone	464	Mask R-CNN	BI: 44.9%, 31.3% PPD: 94.9%, 31.3%	This model locates the picking point better than detecting the bud	Light and wind impact the accuracy	Yan et al. (2022)
BI and PPD	RGB-D camera	1300	YOLOv4	94.19%, 97.29%	New model decreased the size by 75.1% and increased the frame rate by 7.21%	N/A	Guo et al. (2023)
BI	Smart phone	1063	YOLOv7	96.70%,	Model performs well under both strong and weak light conditions	Only tested for one bud with one leaf	Meng et al. (2023)

allowing it to delicately and accurately position the end-effector to pick the tea leaves. These manipulators are combined with AI in order to optimise the motion path. The delta parallel robot (DPR), also known as a three translational degree-of-freedom parallel manipulator, is widely utilised in various applications, particularly in point-to-point pick-and-place operations. This manipulator is favoured for its exceptional accuracy, rapid movement, cost-effectiveness, and high efficiency (Zhang et al., 2020). In tea harvester machinery designs, DPRs are used for locating the end-effector at the correct place of the detected tea buds (Li et al., 2023; Yang et al., 2021).

Expanding the degree of freedom in a manipulator offers numerous benefits. It enhances the manipulator's flexibility, enabling it to access a broader spectrum of positions and orientations in three-dimensional space (M. Zhu et al., 2021a, 2021b). This, in turn, improves the manipulator's reachability, allowing it to traverse areas that may be challenging or inaccessible for a manipulator with fewer degrees of freedom. Additionally, increasing the degree of freedom enhances the manipulator's workspace coverage, accuracy, speed, efficiency, and stability (Nurahmi et al., 2021). Due to these reasons, Wang et al. (2024) conducted a simulation-based investigation on the six translational degree-of-freedom parallel manipulator (6-DOFPM), commonly referred to as a Stewart parallel lifting platform. The findings of this study indicated that the parallel robot exhibited a maximum deviation angle of 10° around the X, Y, and Z axes. It could lift objects up to a maximum distance of 15 cm and had a load capacity of 60 kg. The platform demonstrated a response error within ± 0.1 mm, and its motion characteristics were stable at a sub-millimetre level. The findings validate the suitability of the 6-DOFPM as a viable choice for tea harvesting, presenting it as a feasible alternative to the existing DPR method.

End effectors are the robotic tools or devices attached to the robot's arm or manipulator that are responsible for physically picking or harvesting the tea buds (F. Zhang et al., 2023a, 2023b, 2023c). These end effectors are typically designed to pluck the tea buds without damaging the plant or compromising the quality of the harvested tea. The specific design and functionality of the end effectors used in tea harvesters may vary depending on the tea variety. In designing the end-effector, it is important to consider specific tea leaf properties such as leaf span, leaf width, node spacing, shear force required to break the bud, and the average tea growth region of a bud. This consideration helps prevent damage or picking of non-target buds (Y. Zhu et al., 2021a, 2021b). Additionally, a lightweight design for the end-effector is recommended to minimise the load on the manipulator and increase the overall moving speed.

There are three different types of tea-plucking mechanisms such as Oritsumi, Kakitsumi, and Kokitsumi (Wells et al., 2017). The Oritsumi technique is characterized by grasping

the petioles of selected leaves using the thumb and forefinger, and then plucking them individually by pulling and rotating the hand. The Kakitsumi technique involves a similar method of grasping the leaves, but the hand is pulled upward without any twisting motion. In the Kokitsumi technique, the softer shoots are stripped and pulled out from the bush. Among these techniques, Oritsumi minimises damage to the cut surface, ensuring the preservation of leaves with the highest quality (Motokura et al., 2020; Wells et al., 2017). Manual programming to mimic the hand motion of the Oritsumi technique is very difficult due to its complicated nature. The motion varies depending on factors such as leaf maturity, petiole thickness, and branch length. Therefore, (Motokura et al., 2020) utilised reinforcement learning (RL) and learning from demonstration (LFD) techniques to replicate this motion. The study employed the Jaco robotic arm, which consists of a 6-DOFPM and three fingers as the end-effector. While the motion for picking low and medium stiffness shoots was trained using probabilistic movement primitives (ProMPs), the motion for plucking large stiffness shoots was not trained due to the high force required. Field experiments on tea plucking demonstrated that ProMPs achieved a success rate of 55%, while non-ProMPs achieved a success rate of 45%. This success rate is relatively low compared to other methods used for tea plucking. Therefore, the authors suggest incorporating additional information, apart from shoot stiffness, to train the model and increase the success rate.

Miao et al. (2018) focused on developing a plucking finger that could mimic the manual plucking action of pulling the leaf at the petiole to detach it. To achieve this, they implemented a pneumatic control system to control the plucking fingers. By utilizing genetic algorithms to optimise the geometric parameters of the fingers, they were able to increase the driving force by 26.6%. The researchers reported a plucking success rate of 70% using this design, indicating the effectiveness of the plucking finger in successfully harvesting tea leaves. However, pulling the leaf using a sharp fingertip can damage the tea buds, resulting in a decrease in quality.

In another study, Li et al. (2021) implemented a scissor-type end effector combined with a suction tube in their automated tea harvester design (Fig. 10). The end effector is positioned within one end of the suction tube. When tea plucking occurs, the scissors of the end effector cut the tea shoot, while the suction tube collects the harvested tea leaves and transports them to a collector. Additionally, a negative suction was applied to minimise the errors in positioning the tea bud for the end effector. This negative suction force facilitates the bud to position itself towards the end-effector by slightly bending it (Y. Zhu et al., 2021a, 2021b). The end-effector mentioned was tested to optimise the diameter of the suction tube, the negative suction pressure, and the speed of the manipulator (Y. Zhu et al., 2021a, 2021b). Further, this study analysed the deviation tolerance of positioning the

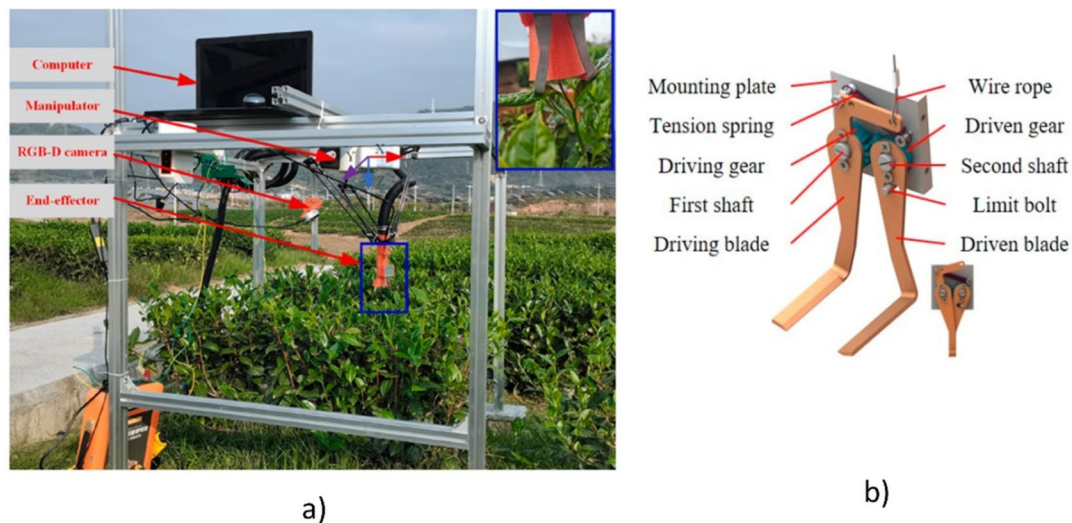


Fig. 10 **a** Automatic tea harvester and enlarged view of the end-effector (Li et al., 2021) and **b** end-effector (Y. Zhu et al., 2021a, 2021b)

suction tube over the tea bud for successful plucking. It was found that the optimal combination of experimental factors consists of a pressure of 0.9 kPa, a diameter of 34 mm, and a velocity of 20 mm/s. By allowing a deviation tolerance range of 10 mm in radius, the average success rate of the end effector's negative pressure guidance is determined to be 97.36%.

Motion Path Planning

Motion path planning for the robotic arm in tea plucking involves determining the optimal trajectory and movement sequence to efficiently and accurately pick the identified tea buds (Xiao et al., 2024). This process considers factors such as the physical constraints of the robotic arm, the positioning of the tea buds, and avoiding any obstacles in the environment (Jin & Han, 2024). The goal is to plan a path that minimises movement time and ensures precise and safe plucking of the tea buds (Li et al., 2023). Robotic arm operations in tea plucking require special considerations due to the dense nature of tea bushes. The movement of the robotic arm must be carefully planned to avoid damaging tea buds and shoots. Additionally, in scenarios where multiple robotic arms are involved, collision prevention between the arms is crucial (X. Zhang et al., 2023a, 2023b, 2023c). To address these challenges, advanced algorithms can be employed to generate motion paths that are both smooth and free from collisions.

In tea plucking, the movement of the robotic arm often takes longer compared to other processes, resulting in reduced harvesting efficiency. Traditionally, a door-shaped arm movement path is used, where the robotic arm reaches a safe height before moving to the next target (Fig. 11a). Since mature tea shoots are located at different heights, the robotic arm needs to move to a safer height (h_i) after harvesting a

shoot and then proceed to the next safe height (h_j) for picking the next shoot. However, using excessively high safe heights (h_{ja}) can increase the overall path length, thereby increasing the moving time (Fig. 11a). To address this issue and shorten the moving time, Li et al. (2023) introduced an improvement by transforming the door-shaped movement path into a parallelogram shape (Oib-Pib-Pjb-Ojb) (Fig. 11b). This modification involved reducing the safe height, effectively decreasing the path length and enhancing the plucking efficiency.

Instead of this door shape motion path, Yang et al. (2021) suggested a “U” shape motion path. This study utilised an improved ant colony algorithm (ACA) for motion planning. In this improved version, a maximum vertical height of the manipulator was set to 10 mm and a secondary trajectory planning (STP) was introduced. For plucking points with small horizontal distance, this STP decrease the path length thus increase the plucking efficiency. The simulation study showed that improved ACA decreased the motion time by nearly 30% compared to unimproved ACA.

Path Sequence Analysis of Robotic Arm.

Path sequence analysis of a robotic arm for tea plucking involves examining the order and sequence of movements performed by the arm to effectively and efficiently harvest tea buds. This analysis aims to optimise the path sequence to minimise time and maximize productivity during the tea-plucking process. In this process, an algorithm identifies the consecutive tea-plucking point with close proximity and schedules the path accordingly. In many applications, the path sequence is predetermined, eliminating the need for real-time analysis. However, in the case of tea harvesting, real-time analysis of the path sequence is crucial.

In a study conducted by Li et al. (2023), a traditional genetic algorithm (TGA) was used to analyse the path sequence (Fig. 11). The robotic arm's path includes the

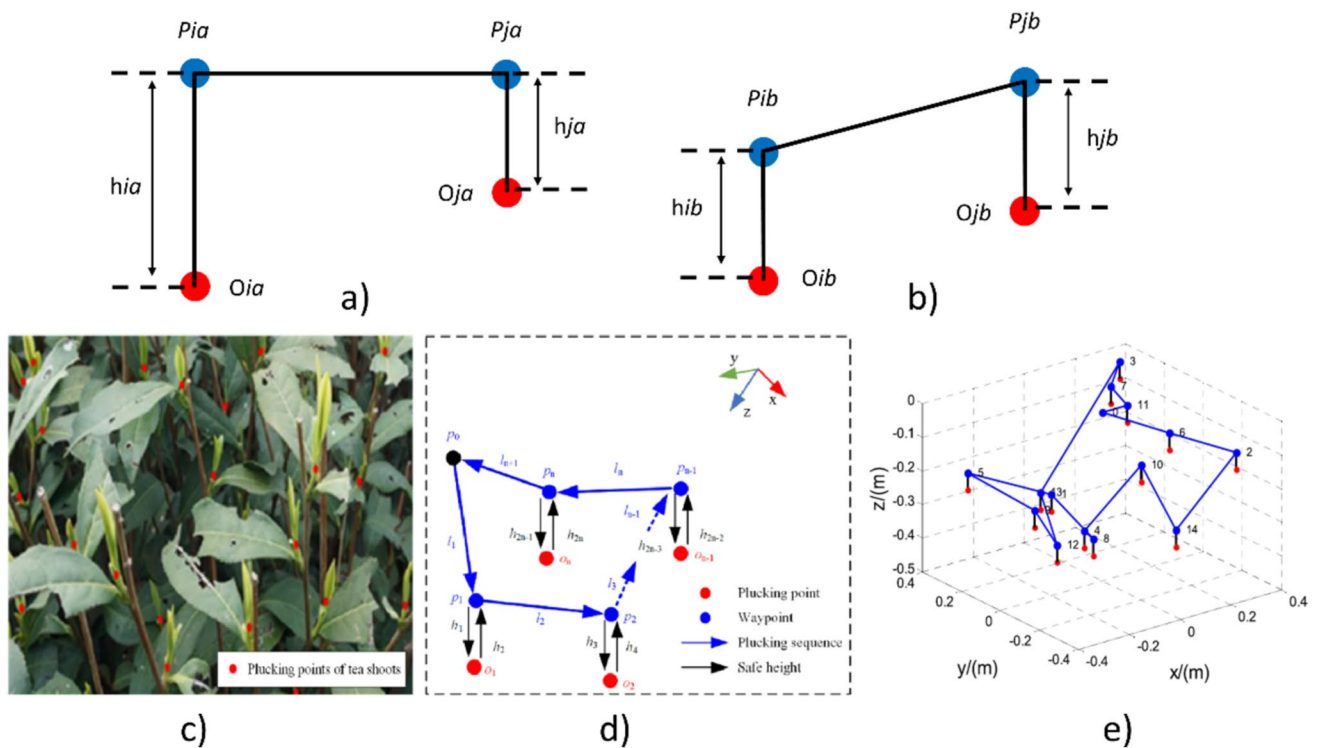


Fig. 11 The plucking path movement and path sequence analysis of robotic arm; **a** door-shape motion, **b** parallelogram-shaped motion, **c** plucking points, **d** 2D trajectory, and **e** 3D trajectory of plucking (adopted from Li et al. (2023))

following steps: first, it moves to a waypoint (safe height); then, it moves to a plucking point; next, it returns to the same waypoint; after that, it moves to a new waypoint; and finally, it moves to a new plucking point. This sequence continues iteratively (Fig. 11d). The total path length is calculated by summing the distance between the safe height waypoints and the distance between tea buds in a 3D trajectory (Fig. 11e). Lin et al. (2023) employed a modified pointer network (PtrNet) that replaces the recurrent neural network (RNN) with a self-attention layer and optimises its parameters using a reinforcement learning algorithm to solve the picking sequence planning problem. The modified pointer network successfully solves the problem with an average path length of 8.30 and an average running time of 1.69 ms for up to 100 tea buds. When compared to other models, namely the genetic algorithm and simulated annealing algorithm, the proposed model was nearly three times faster in solving the path. However, this model follows similar path length compared to other models.

Automatic Tea Harvester Design

A prototype intelligent selective tea harvester was developed by Li et al. (2023). These main components of the harvester are mobile platform, control system, plucking system, and tea leaves collection system (Fig. 12a). The plucking system

is composed of a 3P delta manipulator (a parallel kinematic mechanism with three arms and three prismatic joints) for mounting the end-effector (tea leave plucking component) (Fig. 12 b–d). This was tested in the field for harvesting one bud and one leaf on tea variety was Longjing-43. The success rates for detection, localisation, positioning, and harvesting was 85.16%, 78.9%, 80.23%, and 53.91%, respectively. There are several reasons where failure is identified. Issues with the manipulator include moving beyond the set depth threshold level of the plucking table, limited plucking space for manipulator movement, and exceeding the defined plucking space. Additionally, incorrect estimation of the plucking axis by the algorithm, occlusion of leaves, and incorrect plucking of leaves are also contributing factors to failures in the system.

The average time taken for high-quality point cloud, localisation, and plucking was 0.67 s, 4.88 ms, and 2.17 s, respectively. Out of the process, plucking is the slow process which is the limiting process of the tea harvesting. The average time for the plucking of a shoot was 2.233 s. Tea shoots grow in different level in the Z axis, and therefore, the authors suggest a new manipulator design is required to increase the plucking range. Further, plucking efficiency can be increased by encaging multiple plucking arms. In this study, the camera was placed away from the end-effector. However, positioning the camera at the end-effector is also a possible reason to increase the harvesting accuracy (Gong et al., 2021).

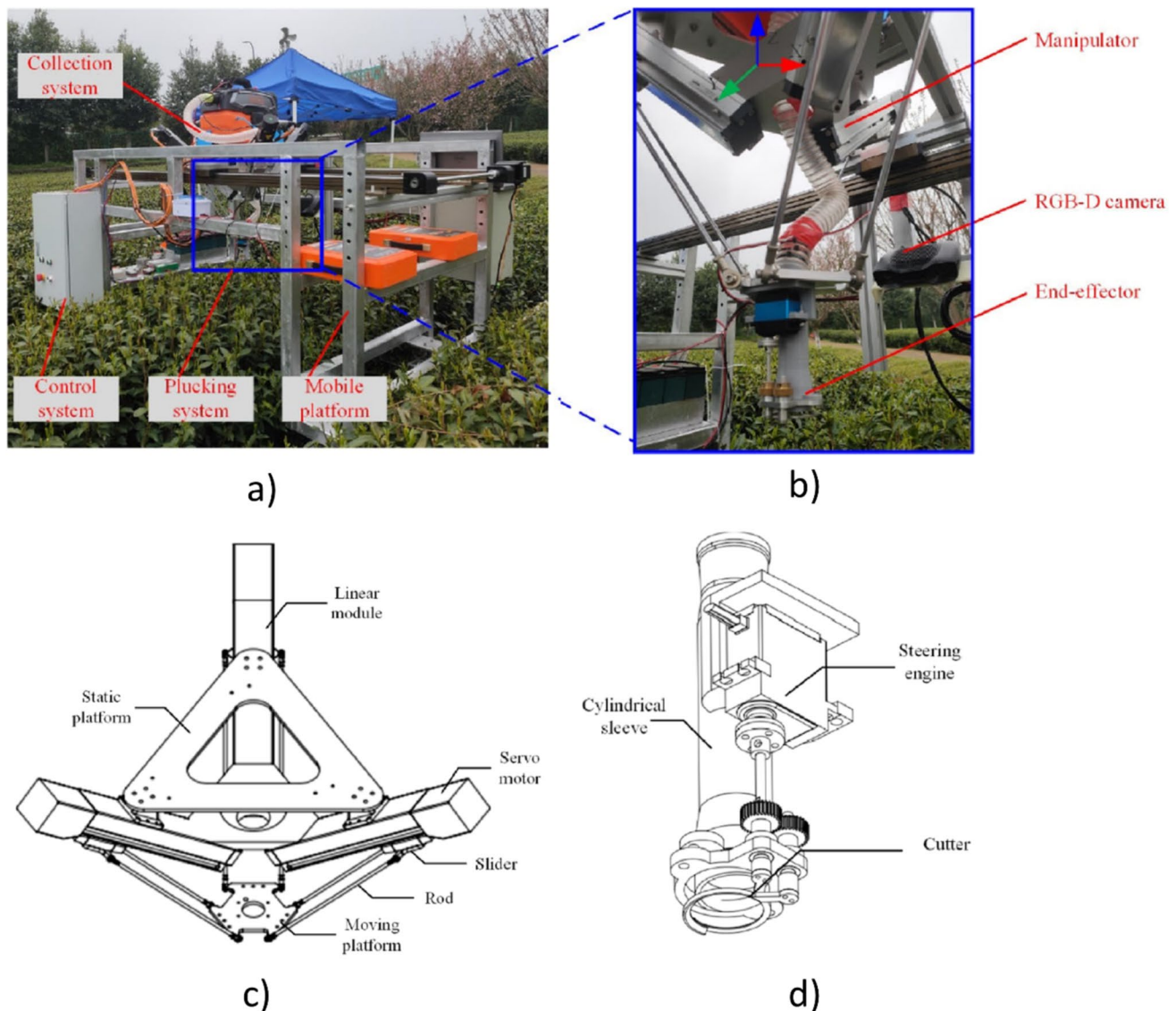


Fig. 12 The automatic selective tea harvester prototype; **a** harvesting robot, **b** plucking arm, **c** 3P delta manipulator, and **d** end-effector (Li et al., 2023)

Future Prospects

Tea plucking is the initial and foremost step in quality tea production. However, conventional manual plucking faces challenges due to a shortage of skilled labour and increasing wages. As a result, mechanical plucking has been introduced to facilitate tea harvesting. Non-selective pluckers are widely used in the tea industry, ranging in size from portable to drivable machines. However, in many older plantations, there is a lack of dedicated and suitable paths for large drivable tea pluckers, leading to the continued use of hand-held shearers. Several attempts have been made to design and fabricate selective tea pluckers, with the “Williames Tea selective plucker” standing out as a remarkable achievement.

This innovative design incorporates two different-sized rollers that enable the selective plucking of tea leaves.

Recently, there has been a growing interest in utilizing computer vision and artificial intelligence in tea harvesting. Research studies have demonstrated the feasibility and reliability of using artificial intelligence-assisted computer vision technology for the selection of tender tea buds. However, the development of an automated tea harvester using this technology is still at the research stage. One significant aspect that consumes a significant amount of time in the sequential steps of tea plucking is the movement of the manipulator. Therefore, it is essential to explore more advanced manipulators and artificial intelligence algorithms to minimise the time required for this task. Additionally, employing multiple

robotic arms for harvesting presents an alternative approach to reduce the overall time and enhance harvesting efficiency.

There is a need for the development of more intelligent tea pluckers that can cater to the specific requirements of different countries and tea grades. Once a computer-assisted tea harvester reaches the commercial level, it should undergo long-term testing under various conditions to evaluate its real impact on tea quality and yield. These intelligent tea pluckers should be made affordable for smallholders to ensure their widespread adoption and application. Considering that most tea fields are not currently suitable for automatic tea harvesters, it is necessary to develop spacing recommendations based on the machinery specifications. Additionally, it is beneficial to reassess existing plucking policies in light of these intelligent pluckers. Further research is warranted to modify these machineries for use in hilly terrains. This aspect is particularly crucial as many major tea-producing countries have tea plantations situated in hilly areas.

Closing Remarks

This study provides a comprehensive overview of tea harvesting machineries and the technological advancements that have occurred in tea plucking. Additionally, this review introduces a comprehensive classification of tea harvesting methods, which can be broadly categorized into four generations based on the evolution of machinery. The first generation of machinery includes simple hand clippers, which have been widely utilised in many countries due to their low cost and the absence of special skill requirements. Recently, mechanical shearers (second-generation machinery) have gained popularity due to their high field coverage and plucking rate. However, these shearers perform non-selective plucking, resulting in decreased tea yield and quality. Large shear plucking machineries are more prevalent in Japan and China. However, these large plucking machinery types are not suitable for many older tea plantations due to a lack of inter-row spacing required for manoeuvring the machinery. Additionally, these machineries are typically wheeled or tracked for convenient use in flat terrains. As a result, they are not as popular in most other countries.

Several attempts have been made to develop mechanical selective pluckers, representing the third generation of tea harvesting machinery. An exemplary design in this category is the “Williames Tea selective plucker”, developed by an Australian company. However, this particular tea plucker has not gained widespread adoption in other major tea-producing countries. Currently, research efforts are focused on computer vision and artificial intelligence-powered machinery for a more precise selection of high-quality tea buds. China is at the forefront of this research, having successfully tested a prototype of such intelligent tea pluckers. However, further improvements are necessary to introduce this technology as a commercial product. One critical aspect that requires scrutiny

is the speed of the manipulator in order to match the efficiency of manual plucking. A major limitation of intelligent pluckers lies in training deep learning models to recognize different types of tea cultivars and to identify various plucking standards for different tea grades. Additionally, research should concentrate on adapting these technologies for use in hilly terrains, reducing production costs to ensure wide accessibility in developing countries, which are major tea-producing nations, and focusing on user adaptation.

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Data Availability No new data was generated in this study.

Declarations

Competing Interests The author declares no competing interests.

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